

So You Caught the Packets. Now What? A Quick Guide to Reading WiFi PCAPs Without Lying to Yourself



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The air in the room is stale. Not metaphorically. Actually stale. Ozone from overworked silicon. Burnt coffee that went cold two refills ago. That metallic laptop fan smell you only notice after hour twelve when your brain starts flattening out.

You did the thing.

You sat in your car in a parking lot you do not belong in. Or hunched over a corner table at a coffee shop pretending to work on a screenplay. High gain

antenna pointed like a divining rod. Wireless card flipped into monitor mode. Capture running.

And now the PCAP is sitting there on your desktop.

A fat file. Quiet. Patient. A little smug.

Digital trophy chest. No instructions inside.

Now what.

Here is the part nobody tells beginners because it ruins the fantasy fast.

Capturing traffic is easy.

Trivially easy.

Monkeys could do it if you duct taped an Alfa card to their hand and installed Kali for them. The tools want to capture. The drivers want to cooperate. The air is already screaming.

The real work starts after. In the silence. In the dim phase where nobody is impressed and nothing blinks green.

This is where operators are made. Or exposed.

So pour another cup of whatever keeps you upright, crack your knuckles, and let's talk about how to actually read WiFi PCAPs instead of worshipping them.

Phase 1: The Sanity Check Everyone Skips Once

Before you fire up Wireshark and cosplay a TV hacker, stop.

Check reality first.

I have watched capable people lose hours because they assumed the capture was valid. Pride is expensive in this field.

Is the file real

Look at the file size. Really look at it.

If you captured for four hours and the file is 24 KB, something is wrong. Antenna unplugged. Wrong channel. Card never entered monitor mode. Capture filter eating everything. Happens more than people admit.

If you do not know what `iwconfig` or `airmon-ng` does, pause. Read. Come back later.

What format did you actually capture

You probably have a `.pcap` or `.pcapng`. If you used `tcpdump`, you may have raw binary that needs decoding.

Wireshark is bloated. Everyone complains. Everyone uses it anyway because it lets you see the horror instead of imagining it.

`tshark` is fine if you hate joy and love terminals. We are assuming Wireshark.

The encryption reality check

This part stings.

If the network is WPA3, you did not capture a handshake or PMKID, and you do not have the password, your PCAP is dead weight.

Encrypted noise. Static. A box of locks without keys.

If it is open, guest, misconfigured, or you caught the handshake, then you actually have something to work with.

Do not skip this step. Hope is not a strategy.

Phase 2: The Firehose Problem

You open the file.

Instant regret.

Thousands of packets. Beacon frames everywhere. Probe requests flying like gnats. QoS data. Retries. Control frames. Management noise.

This is where beginners panic.

Rule one of packet analysis. You do not read everything.

You filter or you drown.

Wireshark filters are not optional. They are survival gear.

Three filters that matter immediately

If you are not filtering, you are watching data vomit.

- `wlan.fc.type_subtype == 0x08`

Beacon frames. Access points announcing themselves. SSID. Encryption type. Capabilities. This tells you what exists before anyone connects.

- `wlan.fc.type_subtype == 0x04`

Probe requests. Devices asking the air if something familiar is nearby. This is where privacy dies quietly. Phones shout old network names like bad confessions.

- eapol

This is the prize. Four packets in sequence from the same MAC. The handshake. Without this, WPA cracking is theater.

If you see it, pause. Smile once. Keep working.

The MAC address dance

Everything revolves around MACs.

BSSID is the access point. Source and destination shift depending on direction. WiFi is weird. Accept it.

Right click a packet. Apply as filter. Selected.

Instant silence. Ninety percent of the noise gone.

This is the moment your breathing slows down.

Phase 3: Reading the Layers Like a Human

Now the fun part. Interpreting intent.

Layer 2: 802.11 reality

This is the WiFi wrapper. Direction flags matter.

To DS equals one and From DS equals zero means client to router.

To DS zero and From DS one means router to client.

Both one means router to router. Rare outside enterprise setups.

Dry. Necessary. Learn it once so you stop misreading conversations.

Layer 3: IP shows the real world

Strip the WiFi skin and suddenly it feels familiar.

Private IPs talking to public ones. TCP. UDP. Structure returns.

Useful filters now:

- `ip.addr == 192.168.x.x` to isolate a single client
- `dns` to read intent in plain text when DoH is not in play

DNS is still the gossip channel. Queries tell stories. Habits. Fear. Curiosity. Sometimes stupidity.

Payloads lie. DNS rarely does.

Phase 4: Cleartext Is Still Everywhere Somehow

This is where cynicism sets in.

Decades of warnings. Entire industries built on encryption. And still people leak everything.

If the network is open or decrypted, start hunting.

HTTP never really died

Filter for `http.`

GET requests everywhere. URLs. Search queries. Admin panels. Analytics beacons.

You may not get passwords often anymore. You still get behavior. Metadata builds profiles better than credentials ever did.

FTP and Telnet still haunt us

Filter `ftp` or `telnet.`

Sometimes you catch an admin managing a relic. Printers. Old servers.
Embedded junk that never learned new tricks.

Credentials in plain text. No ceremony.

It feels unfair. It is not your fault.

IoT talks too much

MQTT. UPnP. Random proprietary chatter.

Smart lights. Locks. Thermostats. Baby monitors.

Payloads often readable. States. Schedules. Presence.

This is not exciting until you realize it maps someone's life better than social media.

Phase 5: WPA2 Decryption Is Work, Not Magic

You caught the handshake. Good.

Now comes reality.

If you know the password, Wireshark can decrypt live. Enable decryption. Add the SSID and PSK. Watch encrypted blobs turn readable.

If you do not know the password, you crack.

Extract handshake. Convert formats. Feed hashcat. Bring GPU.

If you are doing this on integrated graphics, you are not cracking. You are learning patience.

Phase 6: Following the Stream

Packets lie when isolated. Streams tell the truth.

Right click. Follow TCP stream.

Wireshark reconstructs the conversation. Order restored. Meaning emerges.

This is how you find malware callbacks. C2 traffic. Exfiltration patterns. Chat logs. Downloads.

This is where PCAPs stop being noise and start being stories.

The Part Where Watching Turns Into Control

Up to now, you are passive.

Observer. Archaeologist. Cleaner of digital bones.

Then something clicks.

You realize packets are not just caught. They are crafted.

Beacon frames can be spoofed. Probe requests can be answered. Clients can be tricked into trusting ghosts.

This is rogue AP territory.

Not theory. Not lab toys. Real phones connect. Real traffic flows. Real credentials leak.

It is powerful. It is dangerous. It is legally gray in ways that can ruin your year.

If reading this makes your pulse pick up, you are ready for the next layer.

I put together a full guide on building and operating rogue WiFi environments. Hardware, hostapd, dnsmasq, traffic control, social engineering. The boring parts and the sharp edges.

If you want to stop listening and start shaping the air itself, it lives here:

[=>> Grab the Rogue WiFi Access Point Guide here <<==](#)

Use it responsibly. Or at least consciously.

Closing the Capture

Eventually the packets stop.

Wireshark freezes time. The air moves on without you.

PCAP analysis is not just technical skill. It is patience. Pattern recognition. Resisting the urge to jump to conclusions. Seeing humans inside hex.

People checking email. Buying gifts. Making mistakes. Trusting networks they should not.

The air is thick with secrets. Always has been.

Now you know how to listen.

Stay skeptical. Stay curious. Use a VPN on public WiFi. Or better yet, do not connect at all.



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